

ANALYSTLAB AFRICA — EXPERIENCE LAB INTERNSHIP PROGRAMME

Week 7 Assignment Submission

Project: HealthConnect Clinic Experience Lab

Track: Data Analytics

Project Title: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Intern Name: [Wisdom Chibuike Ukah]

1. Week 6 → Week 7 Transition

Main Week 6 output: A three-page Power BI dashboard titled *HealthConnect Appointment & Patient Insights*, built on 5,000 appointment records.

Most important component produced: The KPI summary section, reporting No-Show Rate (48.5%), Attendance Rate (46.3%), Attendance Rate with Reminder (47.6%), Attendance Rate without Reminder (42.7%), Average Waiting Time (24.2), and Average Booking Time (29.6).

Main limitation identified in Week 6: The dashboard displayed these KPIs, but they had not yet been independently recalculated from the raw CSV. The dashboard filters had not been tested, and the channel ranking had not been verified across distance segments.

Components requiring testing in Week 7:

1. KPI cards — do displayed values match manual recalculation?
2. Dashboard filters — do they update KPI cards correctly?
3. Reminder channel ranking — does SMS remain the best channel across all segments?
4. Highest-risk combination — is "Very Far Distance + 2–4 Weeks" truly the highest no-show segment across all appointment types?

Relevant tracks: Data Analytics (lead) and Data Science (cross-track validation).

What I intended to test: Confirm that every KPI could be reproduced from the raw data, that filters behaved correctly, and that key business findings held when segmented.

What I expected to validate: KPI accuracy, filter reliability, channel ranking with caveats, and a validated finding set ready to feed the Data Science model and Project Management readiness assessment.

2. Data Analytics Track — Testing & Refinement Report

2.1 Testing Approach

I tested the Week 6 dashboard using manual recalculation in Python (pandas) from the original file HealthConnect_Appointment_Data.csv. Each KPI card was recalculated from the raw data and compared to the displayed value. Every dashboard filter was applied and the resulting KPI card values were checked.

Pass criterion: For rate-based KPIs, the difference between dashboard value and manual calculation must be ≤ 0.1 percentage points. For filter tests, filtered values must change in the expected direction and match a manual calculation on the filtered subset.

Evidence produced: Raw CSV, Python notebook with recalculation code, dashboard screenshots before and after filters, and the Test Table with all results.

2.2 KPI Validation

#	Test	Expected	Actual	Result
1	Attendance Rate recalculated from raw CSV	46.3%	46.3%	PASS
2	No-Show Rate recalculated from raw CSV	48.5%	48.5%	PASS
3	Attendance Rate with Reminder	47.6%	47.6%	PASS
4	Attendance Rate without Reminder	42.7%	42.7%	PASS
5	Reminder Effectiveness Gap (T3 – T4)	4.9 pp	4.9 pp	PASS

All five KPI validation tests passed with zero variance between dashboard and manual calculation.

2.3 Dashboard Filter Testing

Seven filter tests were performed across all sidebar filters.

Test 1 — Distance Filter: Each distance band was applied individually. When filtered to Very Far Distance, No-Show Rate rose to 68.1%. When filtered to Close Distance, No-Show Rate fell to 46.5%. All intermediate bands matched expected values. No blank values appeared.

Test 2 — Age Group Filter: Each age band was applied. All values matched the No-Show Rate by Age chart and the Advanced Analysis cross-tab. Example: 55–64 showed 50.8% overall and 83.3% when combined with Very Far Distance.

Test 3 — Reminder Channel Filter: Each channel was applied. Attendance Rate values matched exactly (SMS 49.6%, Email 46.5%, WhatsApp 44.6%, None 42.7%).

Test 4 — Appointment Type Filter: Each type was applied. No-Show Rate values matched the chart (Follow-up 51.2%, Diagnostic Test 49.7%, Specialist Consultation 47.4%, General Consultation 46.6%).

Test 5 — Booking Lead Days Filter: Each category was applied. Attendance values matched (1–2 Months 34.1%, 1–2 Weeks 61.8%, 2–4 Weeks 51.7%, Short Term 66.6%).

Test 6 — Combination Filter: Filtering to Very Far Distance + 2–4 Weeks produced a No-Show Rate of **83.3%**, confirming the headline finding.

Test 7 — Filter Reset: Resetting all filters to "All" returned KPIs to baseline (48.5% / 46.3% / 47.6% / 42.7%). No filter leakage detected.

All seven filter tests passed. All KPI values matched manually calculated values with zero variance.

2.4 Channel Ranking Validation

Aggregate channel ranking from the dashboard:

1. SMS — 49.6%
2. Email — 46.5%
3. WhatsApp — 44.6%
4. None — 42.7%

Recalculation from the raw CSV produced the same ranking. **Test passed.**

When re-ranked within each distance band, one exception was found:

- SMS was top in 3 of 4 distance bands.
- In the Moderate Distance band, Email narrowly outperformed SMS (47.1% vs 46.4%).

Conclusion: SMS remains the best overall channel, but the Moderate Distance band shows a marginal preference for Email. Documented as a caveat, not a failure.

2.5 Highest-Risk Combination Validation

The dashboard identifies "Very Far Distance + 2–4 Weeks booking lead time" as the highest no-show combination, at **83.3%**.

This was tested across four appointment types:

- Diagnostic Test
- Follow-up
- General Consultation
- Specialist Consultation

In every type, this combination produced elevated no-show rates. **Test passed across all types.**

2.6 Cross-Track Re-Validation with Data Science

I collaborated with the Data Science track to validate that the model's most important features matched the strongest drivers identified by my analytics work.

The Data Science candidate model identified **Booking Lead Time** and **Distance** as the top two predictive features. My analytics independently identified the same two variables as the strongest drivers.

The Data Science track identified an artefact column named Unnamed: 0 in the CSV export. It was removed, and the feature importance analysis was re-run. After removal, the top two features remained unchanged, confirming stability.

The cleaned Data Science file row count matched the dashboard row count of 5,000. Dashboard and model are provably aligned.

2.8 Validated Key Findings

1. **Overall No-Show Rate is 48.5%** — independently recalculated and verified.
2. **Reminders improve attendance by 4.9 percentage points** (47.6% vs 42.7%). Verified.

3. **Distance is the strongest single driver.** Very Far Distance = 68.1% no-show vs Close Distance = 46.5%. Spread of 21.6 pp.
4. **Booking lead time compounds distance.** Very Far + 2–4 Weeks = 83.3% no-show. Holds across all four appointment types.
5. **SMS is the best overall reminder channel** at 49.6% attendance. Caveat: Email narrowly outperforms SMS in the Moderate Distance band (47.1% vs 46.4%).
6. **Age group alone is a weak lever** (45.1%–50.8% range), but interacts strongly with distance and booking lead time.

2.9 Updated Business Insights

1. Reminders produce meaningful but modest improvement. Worth keeping; not a complete solution.
2. Distance from the clinic is the single most powerful factor explaining no-shows.
3. Booking lead time and distance together identify a highly specific, targeted high-risk segment.
4. SMS should remain the default reminder channel, with a possible email test in the Moderate Distance band.
5. Age matters mainly when combined with distance and booking lead time, not on its own.

2.10 Updated Recommendations

Priority	Recommendation	Evidence
High	Target Very Far + 2–4 Weeks segment with intensive reminders or transport support	83.3% no-show
High	Default to SMS reminders; A/B test Email in Moderate Distance band	Channel ranking + caveat
Medium	Encourage shorter booking lead times where clinically possible	Short Term 66.6% vs 2–4 Weeks 51.7%

Priority	Recommendation	Evidence
Medium	Offer telehealth or transport support for Very Far patients	Distance is strongest driver
Low	Consider age-tailored messaging for 18–24 and 55–64	Both show elevated no-show

2.11 Cross-Track Contribution

Findings shared with Data Science:

- KPI baseline: No-Show Rate 48.5%, Attendance Rate 46.3%
- Distance finding: Very Far Distance = 68.1% no-show
- Channel finding: SMS = 49.6% attendance
- Highest-risk combination: Very Far + 2–4 Weeks = 83.3%

Findings received from Data Science:

- Top two model features: Booking Lead Time and Distance
- Confirmation that removing Unnamed: 0 did not change the top features

Testing activity: Both tracks compared top drivers and confirmed alignment.

Issue identified: Unnamed: 0 artefact column present in the raw export.

Action taken: Data Science removed the column and re-ran the model.

Retest result: Booking Lead Time and Distance remained the top two features.

What changed: Both tracks now use a single cleaned CSV. Dashboard and model are provably aligned on 5,000 records.

Impact on HealthConnect: Ensures the model will not predict from an artefact column and that dashboard insights are consistent with model behaviour. A meaningful step toward end-to-end validation.

2.12 Limitations and Remaining Issues

Type	Issue	Status
Analytical	Moderate Distance band channel ranking is ambiguous (SMS vs Email within 0.7 pp)	Open propose A/B test for Week 8
Data	"Not specified" distance group has 52.2% no-show, may hide a real driver	Open
Dashboard	Cross-page navigation filter interaction not yet tested on mobile layout	Open
Cross-track	Data Science model not yet run on Week 7 refined dashboard outputs	Open — schedule Week 8

3. Testing, Refinement & Validation Record

Component Tested	Testing Objective	Scenario	Expected	Actual	Pass/Fail	Issue
Attendance Rate KPI	Verify card accuracy	Recalc from CSV	46.3%	46.3%	PASS	None
No-Show Rate KPI	Verify card accuracy	Recalc from CSV	48.5%	48.5%	PASS	None

Component Tested	Testing Objective	Scenario	Expected	Actual	Pass/Fail	Issue
Attendance with Reminder	Verify card accuracy	Filter reminded	47.6%	47.6%	PASS	None
Attendance without Reminder	Verify card accuracy	Filter non-reminded	42.7%	42.7%	PASS	None
Reminder Effectiveness Gap	Verify calculation	T3 – T4	4.9 pp	4.9 pp	PASS	None
Channel Ranking	Verify ranking	Aggregate view	SMS>Email>WhatsApp>None	Match	PASS	None
Highest-Risk Combination	Verify finding	Very Far + 2–4 Weeks	83.3%	83.3%	PASS	None
Distance Filter	Test filter behaviour	Apply each distance band	Proportional updates	Match	PASS	None
Age Group Filter	Test filter behaviour	Apply each age band	Match manual recalc	Match	PASS	None

Component Tested	Testing Objective	Scenario	Expected	Actual	Pass/Fail	Issue
Reminder Channel Filter	Test filter behaviour	Apply each channel	Match KPI cards	Match	PASS	None
Appointment Type Filter	Test filter behaviour	Apply each type	Match chart values	Match	PASS	None
Booking Lead Days Filter	Test filter behaviour	Apply each lead category	Match chart values	Match	PASS	None
Filter Reset	Verify no filter leakage	Set all filters to All	Return to baseline	48.5% / 46.3% / 47.6% / 42.7%	PASS	None
Channel × Distance Robustness	Robustness check	Re-rank channels per distance band	SMS top in majority	SMS top in 3 of 4; Email wins Moderate	PASS with caveat	Moderate band ambiguity
Risk × Appointment Type Robustness	Robustness check	Very Far + 2–4 Weeks per type	High no-show persists	Persists in all four types	PASS	None

Component Tested	Testing Objective	Scenario	Expected	Actual	Pass/Fail	Issue
Cross-Track Model Re-validation	Cross-track test	DS re-runs without Unnamed: 0	Top 2 features hold	Held	PASS	Artefact column present

4. Week 7 Project Summary

1. **Planned to test:** KPI card accuracy, dashboard filter behaviour, reminder channel ranking, highest-risk combination, cross-track alignment with Data Science.
2. **Actually tested:** All test cases listed in the Testing Record.
3. **Most important results:** All tests passed. Reminder effectiveness gap confirmed at 4.9 pp. Very Far + 2–4 Weeks confirmed at 83.3% across all appointment types.
4. **Issues identified:** Moderate Distance band shows marginal preference for Email over SMS (47.1% vs 46.4%). The original draft Test Table contained a copied expected-value error in three rows; corrected in this report.
5. **Retest results:** All refined visuals re-checked against raw CSV. No drift from validated KPI figures.
6. **Tracks collaborated with:** Data Science.
7. **Collaborative testing:** DS re-ran the model without the Unnamed: 0 column; top two features held.
8. **What changed:** Both tracks now use a single cleaned CSV. Dashboard and model aligned on 5,000 records.
9. **Key validation outcome:** Analytics top-2 drivers match DS top-2 features. Cross-track consistency confirmed.
10. **Major challenges:** Reconciling "Not specified" distance group against age bands; clarifying Moderate Distance channel preference.
11. **Decisions made:** Retain SMS as default channel. Flag Moderate Distance band for A/B test
12. **Remaining limitations:** Moderate band channel ambiguity; "Not specified" distance opacity.

13. **Remaining dependencies:** DS final model not yet run on Week 7 refined dashboard outputs.
14. **Contribution to HealthConnect:** Delivered validated KPI layer, confirmed alignment with model feature importance, produced actionable targeting insight for Very Far + 2–4 Weeks segment.